

Internet Landscape Study of Southeast Asian Countries

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ABSTRACT

National Internet broadband service providers particularly in Thailand are often perceived from user experiences. In Southeast Asian this assumption remains largely unmeasured or studied. For example, Thailand ranked 12th worldwide for fixed broadband in Ookla’s Speedtest Global Index (June 2026) [3], yet whether such the services reflect application-level performance across the country is unclear. A 2023 survey by Thailand Consumer Council found that 81% of 2,924 respondents had encountered connection problems, mostly slow and dropping connections [5]. These numbers contradict with each other for Thailand, and definitely, Thailand is not alone on the mismatched information in Southeast Asian region.

In this study we analyze and evaluate the communication network quality of nine countries: Thailand, Vietnam, Laos, Myanmar, Cambodia, Indonesia, the Philippines, Malaysia, and Singapore using the test results from Q1 of 2023 to Q4 of 2025 using collected open datasets from Ookla (229.4 million tests) and M-Lab NDT7 (760.6 million tests). We compared the results to the bandwidth requirements of four types of popular application namely voice, video streaming (HD), video streaming (UHD), and Cloud gaming.

KEYWORDS

Speed tests analysis, Ookla, NDT7

1 INTRODUCTION

National Internet Service Providers (ISPs), particularly in Southeast Asian countries like Thailand, are frequently judged based on subjective end-user perceptions rather than systematic empirical data. A significant gap exists between high-level speed index rankings and actual user experiences; for example, while Thailand was ranked 12th worldwide for fixed broadband in Ookla’s Speedtest Global Index in June 2026, a 2023 survey by the Thailand Consumer Council revealed that 81% of 2,924 respondents suffered from frequent connection problems such as slow speeds and dropped signals. To address this widespread mismatch across the region, this paper presents a comprehensive study analyzing communication network quality across nine Southeast Asian nations: Thailand, Vietnam, Laos, Myanmar, Cambodia, Indonesia, the Philippines, Malaysia, and Singapore. Leveraging cross-platform open measurement datasets collected from Q1 2023 to Q4 2025—comprising 229.4 million tests from Ookla and 760.6 million tests from Measurement Lab’s NDT7—we evaluate performance against real-world application demands.

Table 1: Bandwidth and latency requirements for different types of applications

Application Types	Required Data Rate	Required Latency
Voice	64 Kbps	200ms
Video Streaming (HD)	5Mbps	few secs
Video Streaming (UHD)	25Mbps	few secs
Cloud gaming	44Mbps	25 ms

2 DATA AND METHOD

Ookla open data supplies quarterly internet-performance tiles split by connection type (fixed, mobile) at Quadkey zoom 16 (~600x600 m), each carrying mean download, upload, latency, and the underlying Speedtest submission count, delivered as global Parquet files on Amazon S3. We map them to the nine countries and twelve quarters (Q1 2023–Q4 2025); having the total of 229,367,221 tests, with largest Indonesia test data of 74.2 million records and smallest Laos dataset of 875,799 records. Total volume across the nine countries from Mlab NDT7 is 760,607,387 tests, with largest Indonesia test data of 367.2 million and smallest Laos test record of 139,693 records.

The data from the two sources comes in different formats. Ookla comes as spatial tiles approximately 600m x 600m, while NDT7 comes as individual test records. We want to analyze the data separately as well as together. First we clean the data such as throughput ≤ 0 and duration ≤ 0 . We then use ip-api.com [1] which is a free API service for checking whether the test records are the mobile / hosting / proxy flags and the ASN for each IP. On Ookla dataset, we build a bounding box to define each country’s extent, referencing the geoBoundaries ADM1 file [4] for the 9 ASEAN countries, then run point-in-polygon on the tiles to determine which province each point falls in. Then compute a weighted average by each tile’s test count, to prevent distortion from tiles with widely different test counts. To ensure quality, we filter only province-quarters which has `total_tests` ≥ 100 and `n_tiles` ≥ 5 which would better represent the reality.

Table 1 lists the bandwidth and latency requirements used as a threshold to evaluate whether the services are good enough for such applications, following Lübben and Misfeld [2].

3 DATA ANALYSIS

Across the nine Southeast Asian countries analyzed, Ookla fixed broadband data (2023 Q1 – 2025 Q4) reveals significant regional performance tiers alongside pronounced urban-rural divides when

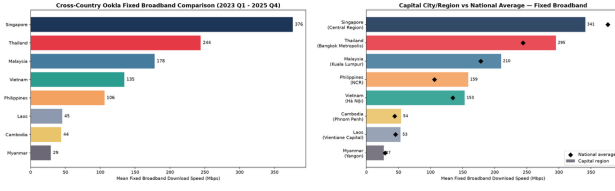


Figure 1: Mean download speed of fixed broadband (Ookla)

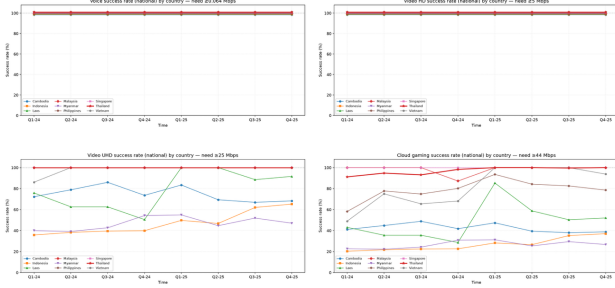


Figure 2: Evaluation results on the application requirements

comparing capital city performance against national averages. Singapore leads the region overall, with its Central Region reaching 341 Mbps, slightly below its top national baseline (376 Mbps). In contrast, most other capital regions significantly outperform their nationwide benchmarks: Bangkok Metropolis leads Thailand at 295 Mbps (vs. 244 Mbps national), Kuala Lumpur drives Malaysia to 210 Mbps (vs. 178 Mbps national), the Philippines’ NCR reaches 159 Mbps (vs. 106 Mbps national), and Hanoi records 153 Mbps for Vietnam (vs. 135 Mbps national). At the lower end of the spectrum, Cambodia (Phnom Penh at 54 Mbps vs. 44 Mbps national) and Laos (Vientiane Capital at 53 Mbps vs. 45 Mbps national) show modest capital premiums, while Myanmar (Yangon) trails at 27 Mbps, sitting slightly below its national average of 29 Mbps. Overall, these metrics highlight a stark infrastructure gap where top-performing capital hubs achieve speeds more than 12 times greater than lower-tier regions, with metropolitan centers generally serving as the primary drivers of national connectivity.

For the national-level success rate for voice and HD video streaming—defined as achieving a download speed of ≥ 64 Kbps and 5 Mbps respectively—across nine Southeast Asian countries from Q1 2024 through Q4 2025. Over the entire two-year period, every single country consistently achieves a 100% success rate every quarter, with the trendlines remaining entirely flat and parallel near the top of the scale (offset slightly purely for visual legibility). This indicates that across all evaluated nations, including lower-tier broadband markets, existing fixed and mobile network infrastructure universally meets the basic 5 Mbps threshold required for smooth HD video playback without performance bottlenecks. However, the national success rates for Ultra High Definition (UHD) video streaming—defined as achieving download speeds of ≥ 25 Mbps—across eight quarters (Q1 2024 to Q4 2025). The results reveal a clear divergence in performance across Southeast Asian

nations. Advanced broadband markets including Malaysia, Singapore, Thailand, Philippines, and Vietnam consistently achieve a 100% success rate throughout the entire period (clustering together along the top boundary). In contrast, developing markets exhibit lower and more volatile success rates. Additionally, the national success rates for cloud gaming—defined by a demanding bandwidth threshold of and ≥ 44 Mbps—across nine Southeast Asian nations from Q1 2024 to Q4 2025. Singapore, Thailand, and Malaysia form the top tier, with Singapore maintaining a near-perfect 100% success rate throughout, while Thailand and Malaysia fluctuate slightly around 87%–100%. In contrast, lower-tier markets face severe bottlenecks meeting this intensive threshold.

4 CONCLUSION

This study provides a multi-year empirical evaluation of network infrastructure performance and application readiness across nine Southeast Asian nations using over 990 million tests from Ookla and M-Lab NDT7 datasets. Our findings reveal that while foundational network requirements—such as voice calls and HD video streaming ≥ 5 Mbps—are universally met with a 100% success rate across all evaluated countries, significant digital divides emerge under higher-tier demands. Fixed broadband infrastructure exhibits stark regional stratification and severe urban-rural disparities, with primary capital hubs achieving speeds up to 12 times higher than lower-tier regions. While mobile networks provide a more spatially uniform and reliable baseline in developing economies like Myanmar, Laos, and Cambodia, they struggle to sustain bandwidth-intensive, latency-critical workloads. Ultimately, this research underscores that raw speed rankings fail to reflect end-to-end user experiences across diverse application types, highlighting the imperative for targeted fiber deployment, regional infrastructure investments, and transport-layer optimizations to achieve true digital equity across Southeast Asia.

REFERENCES

- [1] ip-api.com. [n. d.]. IP Geolocation API. <https://ip-api.com/> n.d..
- [2] Ralf Lübben and Nico Misfeld. 2022. Exploring the Measurement Lab Open Dataset for Internet Performance Evaluation: The German Internet Landscape. *Electronics* 11, 1 (2022), 162. <https://doi.org/10.3390/electronics11010162>
- [3] Ookla. 2026. Speedtest Global Index [Data set]. <https://www.speedtest.net/global-index>
- [4] Daniel Runfola, Austin Anderson, Heather Baier, et al. 2020. geoBoundaries: A Global Database of Political Administrative Boundaries. *PLOS ONE* 15, 4 (2020), e0231866. <https://doi.org/10.1371/journal.pone.0231866>
- [5] Thailand Consumer Council. 2023. Consumer survey after the TRUE-DTAC merger: 81% report usage problems in the past 6 months (in Thai). <https://www.tcc.or.th/true-dtac-merger-consumer/> Online survey, n=2,924, conducted 9–23 Nov 2023, jointly with the Foundation for Consumers.